Bholanath Bala

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About Me

Electronics and Communication Engineering graduate with strong interests in mathematical modeling, machine learning, and data science. Experienced in transformer-based deep learning models and computational analysis, with a motivation to apply mathematical and computational approaches to complex problems in artificial intelligence and intelligent systems.

Education

Bachelor of Science (B.Sc.) in Electronics and Communication Engineering (ECE).
Khulna University, Bangladesh
CGPA: 3.57/4.00 | **Graduated:** 2026

Research Experience

Undergraduate Thesis: A Comparative Study of Vision Transformers and Time-Series Transformers for Bearing Fault Diagnosis Using PU Dataset.

- Developed transformer-based deep learning models for automated machinery fault diagnosis.
- Implemented and evaluated **Vision Transformer (ViT)** and **Time-Series Transformer (PatchTST)** architectures.

Research Publication

1. ***"Vision Transformer-Based Classification of Bearing Faults from Vibration Signal Spectrograms."*** Paul, A., Bala, B., Hasan, M. N., & Toma, R. N. (2025).
    - **2025 28th International Conference on Computer and Information Technology (ICCIT),** pp. 2796–2801. IEEE.
    - **DOI:** 10.1109/ICCIT68739.2025.11490241
    - **IEEE Xplore:** https://ieeexplore.ieee.org/document/11490241
2. ***"A Novel Attention-Enhanced Hybrid Deep Learning Model for GI Disease Classification."*** Maniruzzaman, M., Salman, R. I., Tawhid, K. G., Bala, B., Hossain, M. M., & Ahsan, M. S. (2026).
    - **2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII).** IEEE.
    - **DOI:** 10.1109/PECCII70991.2026.11661820
    - **IEEE Xplore:** https://ieeexplore.ieee.org/document/11661820

Academic Projects

1. **Monitoring Environmental Conditions for Sundari Trees in the Sundarbans.**
    - Developed an **IoT-based environmental monitoring system** using sensors to monitor environmental conditions.
    - Designed web and mobile interfaces for real-time monitoring and automated alerts.

- Enabled remote access to environmental data for improved monitoring and decision-making.

2. **Combating Unpaid Food Purchases Using OpenCV and Face Recognition**
    - Developed a computer vision system using **Python and OpenCV** to identify unpaid food purchases.
    - Applied face recognition techniques to improve automation and security in canteen environments.
3. **University Certificate Automation System**
    - Developed an automated system for managing student certificate approval workflows.
    - Designed and implemented a responsive interface using **React.js**.
    - Implemented state-driven UI and real-time status tracking.

Technical Skills

- **Programming:** Python, C, Java, JavaScript, TypeScript, MATLAB
- **Web Technologies:** HTML, CSS, Tailwind CSS, React.js
- **Data Science & ML:** NumPy, Pandas, Matplotlib, Machine Learning, Deep Learning
- **Tools:** Git, LaTeX

Achievements

- **Champion — Research Category, Smart Bangladesh Seminar,** Khulna University, for the Sundarbans Environmental Monitoring Project.
- **2nd Runner-up — Machine Learning Poster Design Competition,** Khulna University.
- **Winner — Open Doors Russian Scholarship Project 2025/2026 (Olympiad),** Applied Mathematics and Artificial Intelligence.

Training & Experience

- **Industrial Training in Data Science at Grameenphone, Bangladesh.**
    - Gained foundational knowledge in data analysis and machine learning workflows through industry-led training sessions

Certifications

- Programming for Everybody (Getting Started with Python), Coursera
- Intermediate Python, DataCamp
- Introduction to Machine Learning, Kaggle
- Introduction to SQL, DataCamp
- Introduction to Git, DataCamp

Languages

- **Bangla:** Native
- **English:** Intermediate

Recommendations

1. **Dr. Rafia Nishat Toma**
   Associate Professor
   Electronics and Communication Engineering (ECE) Discipline

Khulna University, Khulna-9208, Bangladesh
Email: rafiatoma@ece.ku.ac.bd
Relationship: B.Sc. Thesis Supervisor

2. **Dr. Md. Maniruzzaman**
Professor
Electronics and Communication Engineering (ECE) Discipline
Khulna University, Khulna-9208, Bangladesh
Email: m_m_zaman@ece.ku.ac.bd